

Expanding the Mathematical Foundations of Artificial Life: A Systematic Search Across Function Families, Geometries, and Open Expressions in Lenia

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Abstract

Lenia, a continuous cellular automaton, has demonstrated rich lifelike behaviors within a narrow family of hand-designed kernel and growth functions. We systematically expand Lenia’s search space along three axes: function families (from 4 to 28 kernels and 4 to 16 growth functions), geometric substrates (flat, spherical, toroidal, hyperbolic), and open mathematical expressions generated by a large language model (LLM). Using MAP-Elites quality-diversity search across 125,000+ evaluations, we illuminate 291 behavioral niches. Our central finding is that LLM-guided search discovers $\cos(4\pi \cdot \nabla^2 A)$ modulating the growth center — a curvature-coupled term that single-kernel Lenia cannot express (its unit-normalized kernels exclude the zero-mean Laplacian operator) and that the additive multi-kernel form $\sum_i G_i(K_i * A)$ cannot express either, placing it outside the function class that standard parametric search explores. A three-track ablation (parametric, LLM-guided, random baseline) decomposes contributions from search space richness, inductive bias, and LLM precision tuning. Beyond extended textures, targeted localization search discovers 50 moving solitons from 17 kernel/growth combinations — 49 from expanded function families, one from an original kernel paired with a new growth function — with anisotropic kernels opening wider locomotion basins reachable by coarse-grained search. The most robust creature, a `bessel/gaussian_bell` soliton, achieves kernel-determined autonomous motion (9/9 survival, direction std = 2.1°). The frontier of artificial life lies in expanding the mathematical space being searched, not only the search algorithm.

Introduction

The search for artificial life has historically been shaped by the mathematical structures researchers choose to explore. Lenia (Chan, 2019) generalizes Conway’s Game of Life to continuous states, space, and time, with the update $A_{t+1} = \text{clip}(A_t + dt \cdot G(K * A_t), 0, 1)$. Within this framework, Chan discovered over 400 species of “digital

organisms” — localized, self-sustaining patterns with internal structure and autonomous motion. But these discoveries were made within a narrow mathematical corridor: Gaussian kernels and bell-shaped growth functions. This raises a fundamental question: **what other mathematics can produce life?**

We approach this question through systematic expansion along three dimensions:

Function families. We expand from 4 kernel functions to 28 (adding functions from mathematical physics, signal processing, and approximation theory) and from 4 growth functions to 16 (adding ecological models and dynamical systems). Each family encodes different assumptions about spatial interaction and growth dynamics.

Geometric substrates. We simulate Lenia on the flat plane, sphere S^2 , torus T^2 , and hyperbolic plane H^2 , altering the topology of spatial interactions from bounded to periodic to exponentially expanding.

Open expressions. Moving beyond parametric templates, we use a large language model (Claude, Anthropic) to generate arbitrary mathematical expressions as Lenia update rules, combining convolution, Laplacian, gradient, and other differential operators unconstrained by human-designed function families.

To search this expanded space, we employ MAP-Elites (Mouret and Clune, 2015), a quality-diversity algorithm maintaining an archive indexed by pattern complexity (LZMA compression ratio) and spatial density (alive fraction) over a 20×20 grid. Our key findings are: (1) LLM-guided search discovers $\cos(4\pi \cdot \nabla^2 A)$, a curvature-coupled growth term outside single-kernel and additive multi-kernel Lenia, producing geometric patterns no standard parametric search reaches; (2) a three-track ablation decomposes contributions from search space, inductive bias, and LLM guidance; and (3) targeted localization search discovers 50 moving solitons from 17 kernel/growth combinations, 49 from expanded function families and one from an original kernel with a new growth function, progressing from self-

organizing textures to stationary solitons to autonomously moving creatures, with anisotropic kernels widening the reachable locomotion basins.

Related Work

Lenia and Continuous Cellular Automata

Lenia (Chan, 2019) introduced continuous cellular automata with update $A_{t+1} = \text{clip}(A_t + dt \cdot G(K * A_t), 0, 1)$. Chan (2020) expanded this to multi-channel systems, and Flow-Lenia (Plantec et al., 2023) addressed mass conservation and parameter localization. A parallel line removes Lenia’s clip to expose its continuous dynamics: Asymptotic Lenia (Kawaguchi et al., 2021) and its reaction-diffusion realization (Kojima and Ikegami, 2023) cast Lenia as a kernel-based Turing system, while mass-conserving variants (Flow-Lenia; MaCE (Papadopoulos and Guichard, 2025)) transport mass along affinity gradients. These works demonstrated rich lifelike behavior but explored a limited portion of the possible mathematical space. Our LLM-found $\cos(4\pi\nabla^2 A)$ growth is best read against this backdrop: a cosine of curvature selects a spatial wavelength, the mechanism underlying classical pattern-formation PDEs (Swift–Hohenberg (Swift and Hohenberg, 1977), Cahn–Hilliard (Cahn and Hilliard, 1958)). Our contribution is not a new PDE but the finding that open, search-driven exploration lands on this curvature-coupled regime — one the standard single-kernel parameterization cannot reach.

Automated Search in Artificial Life

Leniabreeder (Faldor and Cully, 2024) applies MAP-Elites to discover diverse Lenia patterns, searching *within* the standard functional family using numeric parameters. Our Track A subsumes this parametric search (28 kernels $\times$ 16 growth functions $\times$ 4 geometries) and Track B searches over symbolic expressions. **ASAL** (Kumar et al., 2024) uses foundation models as *evaluators* of visual novelty across five substrates. The critical distinction is that ASAL’s search never escapes the standard parameterization, while we use the LLM as a *generator* of novel mathematical expressions — our central finding concerns structures that lie *beyond* the parametric framework. ASAL’s multi-substrate coverage versus our Lenia-only focus represents a scope-depth trade-off; our three-track ablation provides controlled decomposition that ASAL does not include.

LLMs for Scientific Discovery

LifeGPT (Berkovich and Buehler, 2024) trains a transformer to *predict* known cellular automaton dynamics; in contrast, we use LLMs to *generate* novel update rules. FunSearch (Romera-Paredes et al., 2023) and AlphaEvolve (Google DeepMind, 2025) demonstrated that LLM-generated code can surpass human-designed algorithms. We apply this paradigm to artificial life, where LLM-generated expressions serve as update rules evaluated by emergent behavior.

Our Track C random baseline isolates the LLM’s specific contribution as precision tuning of mathematical structures.

Method

Lenia Framework

We adopt the standard Lenia update rule:

$$A_{t+1} = \text{clip}(A_t + dt \cdot G(K * A_t), 0, 1) \quad (1)$$

where $A_t \in [0, 1]^{N \times N}$ is the state field, K is a radially symmetric convolution kernel parameterized by a shape function $k(r)$ and radius R , G is a growth mapping, and dt is the time step. We retain the clip to $[0, 1]$ throughout: it bounds the state and keeps the discovered rules directly comparable to standard Lenia, whereas removing it yields the asymptotic/PDE formulations (Kawaguchi et al., 2021; Kojima and Ikegami, 2023) we leave to future work.

Expanded Function Families

We expand from 4 to **28 kernel functions** across five categories: classic (original Lenia), oscillatory (Bessel, sinc, chirp), physics-inspired (Lennard-Jones, Morse, Riesz, Yukawa), windowed (Wendland C2, sech), and geometric (power-law, elliptical, Matérn). Similarly, we expand from 4 to **16 growth functions**: classic, heavy-tailed, ecological (Ricker, Allee), dynamical systems (FitzHugh-Nagumo, Swift-Hohenberg), and wavelet-inspired. All kernels are normalized to integrate to 1; all growth functions output $[-1, 1]$. Complete definitions are in Appendix A–B.

Geometric Substrates

We simulate Lenia on four substrates: flat plane ($\mathbb{R}^2$, periodic boundary), sphere (S^2 , geodesic distance with $\cos \theta$ area correction), embedded torus (T^2 , non-uniform curvature), and hyperbolic plane (H^2 , Poincaré disk with exponentially growing neighborhoods).

Track A: Parametric MAP-Elites Search

Track A searches $\{28 \text{ kernels}\} \times \{16 \text{ growth}\} \times \{4 \text{ geometries}\} \times \mathbb{R}^5$, with continuous parameters sampled uniformly from kernel radius $R \in [5, 30]$, peaks $p \in [0.5, 4]$, growth center $\mu \in [0.05, 0.5]$, width $\sigma \in [0.005, 0.15]$, and time step $dt \in [0.01, 0.5]$. Each evaluation is seeded with uniform noise in a central patch. Mutation perturbs 1–2 parameters: categoricals resampled uniformly, continuous parameters via Gaussian noise of 10–30% of range (clipped). The seeding 287-entry archive is itself from earlier staged Track A runs (4 kernels, then the 15-kernel expansion, then four geometries).

Track B: LLM-Generated Expressions

Track B uses Claude (Anthropic) to generate mathematical expressions compiled into PyTorch computation graphs. Available operations include `conv`,

laplacian, gradient, sin, cos, tanh, exp, with additional high-level operations for dual-field mode (curl, compete, conserve, rotate; full list in the released code). The Laplacian and gradients are fixed finite-difference stencils (the standard 5-point discrete Laplacian $[[0, 1, 0], [1, -4, 1], [0, 1, 0]]$ with periodic boundary), applied identically across all expressions. Expressions are validated against an operation whitelist with per-step timeout. The evolutionary loop generates initial expressions, then iteratively mutates by combining parents from the archive.

Behavior Space and Archive

Both tracks share a MAP-Elites archive indexed by two behavior descriptors of the final state — complexity c and alive fraction a — discretized into a 20×20 grid (400 bins). Complexity is the LZMA compression ratio of the final field quantized to 8 bits ($\lfloor 255A \rfloor$, lzma preset 1), a practical substrate-agnostic proxy for Kolmogorov complexity (Zenil et al., 2018); alive fraction is the fraction of cells with $A > 0.01$. The per-cell **fitness is the complexity c itself**, and only the highest-complexity entry per cell is retained, with all tracks competing for the same cells. Using a descriptor as the fitness is a deliberate simplification: it provides only weak within-cell selection (toward the high-complexity edge of each complexity bin), so the archive is best read as a diversity-illumination map (see Limitations); the cross-track *quality* comparisons (Tables 3, 5) re-evaluate entries on complexity under identical conditions.

Evaluation Protocol

Track	Resolution	Steps	Evals	Seeded
A (parametric)	128^2	8,000	$\sim 125\text{K}$	287
B (LLM)	64^2	2,000	$\sim 5\text{K}$	287
C (random)	64^2	2,000	1,000	None
Locomotion	256^2	10,000	$\sim 15\text{K}$	None

Table 1: Evaluation protocol for each search track.

The locomotion search uses anisotropic kernel perturbation ($K \leftarrow K \cdot (1 + \varepsilon \cos(\theta - \theta_0))$, $\varepsilon \in [0.1, 0.6]$) and fitness = concentration $\times$ net_displacement. Track A ran on $4 \times \text{H100}$ GPUs (~ 5.5 hours, $\sim \$43$); Track B locally (~ 8 hours, $\sim \$15$ API). Key entries are re-evaluated under identical conditions (128^2 , 8,000 steps, 10 seeds) for fair comparison (Section 4.3). Computational details (hardware, run-times, and costs) are in the released repository.

Results

Archive Coverage

The unified archive covers **291/400 bins (72.8%)**, with Track A contributing 285 entries and Track B contributing 6 (Figure 1). Empty bins concentrate in the high-complexity, low-alive-fraction region.

Structural Expressiveness Beyond Parametric Frameworks

Track B discovers a growth term — $\cos(4\pi \cdot \nabla^2 A)$ modulating the growth center — that lies outside the function class explored by standard parametric Lenia. We state this precisely at two levels. *Single-kernel Lenia* applies a single-argument growth G to one convolution $u = K * A$, so G cannot depend on any spatial statistic of A not carried by u ; the hero term makes growth depend additionally on the Laplacian $\nabla^2 A$. One cannot simply set $K = \nabla^2$: the discrete Laplacian kernel is zero-mean and sign-indefinite, whereas Lenia kernels are non-negative and normalized to $\int K = 1$ (Method), so it is not an admissible kernel. *Additive multi-kernel Lenia* (Chan, 2020) sums one-argument growths $\sum_i G_i(K_i * A)$, and so cannot reproduce the hero’s *nonlinear* coupling, in which $\nabla^2 A$ shifts the center of a Gaussian bell evaluated at $\text{conv}(A)$. The claim is thus scoped to these standard parametric families; it does *not* hold for multi-channel Lenia (Chan, 2020) or Neural Cellular Automata (Mordvintsev et al., 2020), which expose several fixed convolutions (Laplacians included) to a joint cross-channel map and can represent such coupling — our point is reachability, not absolute impossibility. Mechanistically, the cosine of the Laplacian creates zero crossings that discretize the density field into periodic stable bands, producing geometric self-organization with no analogue in our parametric archive. The full expression and component analysis are in Appendix C.

A manual verification confirms that the Laplacian-modulated growth center alone (implementing only $\mu = \mu_0 + \varepsilon \cos(4\pi \nabla^2 A)$ without the bandpass window or harmonic modulation) produces Turing-like spot patterns (complexity 0.13) rather than the hero’s geometric rings (complexity 0.94). The full crystallographic pattern emerges from the interaction of all three components (Appendix C), indicating that the LLM’s contribution includes not only the structural innovation of spatially-aware growth but also the precise arrangement of supporting nonlinearities.

The expression produces highly consistent patterns across 20 runs at two resolutions (coefficient of variation $< 0.3\%$):

Resolution	Alive frac.	Complexity	Clusters	Entropy
128^2 ($n=10$)	0.977 ± 0.001	0.939 ± 0.002	6.1 ± 1.6	3.49 ± 0.11
256^2 ($n=10$)	0.983 ± 0.001	0.899 ± 0.002	29.3 ± 2.7	3.04 ± 0.08

Table 2: Hero expression robustness across resolutions.

Of Track B’s surviving entries, 88% employ high-level operations; prompt strategy analysis is in the released repository.

Controlled Comparison

Under identical conditions (128^2 , 8,000 steps, 10 seeds each), Track A dominates the fitness ranking; the hero ranks

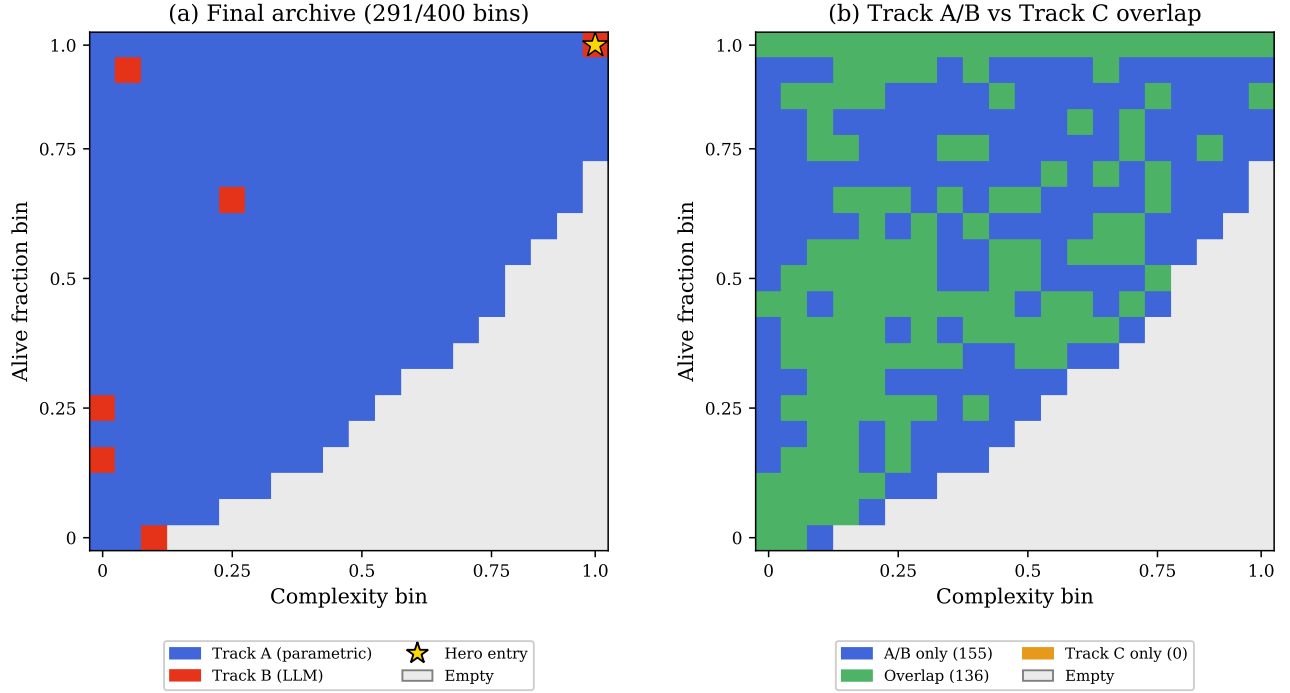


Figure 1: Behavior space archive map. (a) Final archive with 291/400 bins. Track A (blue), Track B (red), hero entry (star). (b) Track A/B vs Track C overlap: green = shared bins, blue = A/B exclusive.

sixth on complexity (Table 3), so its significance lies in the novel structure it realizes rather than in peak fitness:

Rank	Track	Entry	Mean complexity $\pm$ std
1	A	sech / FitzHugh-Nagumo	0.998 ± 0.001
2	A	sech / FitzHugh-Nagumo	0.971 ± 0.001
3	A	Yukawa / Laplace peak	0.960 ± 0.004
4	A	Riesz / Gaussian bell	0.951 ± 0.004
5	A	power-law / bistable	0.950 ± 0.008
6	B	$\cos(4\pi \cdot \nabla^2 A)$	0.939 ± 0.002

Table 3: Top 6 entries under controlled re-evaluation (128^2 , 8,000 steps, 10 seeds).

Morphological Divergence

Track A produces organic, irregular morphologies (worm-like textures, foam, spirals) characteristic of reaction-diffusion systems. The hero expression produces a categorically different pattern (Figure 2): concentric waves with crystallographic regularity, arising because the cosine discretizes the Laplacian field into periodic stable bands — a mechanism unavailable to single-kernel parametric growth, which receives only a scalar convolution value.

Function Family Analysis

Of 286 Track A entries, the original 4 Lenia kernels plus the 11 first-expansion kernels (the first 15 in our kernel enumeration) account for 82%, while the 13 kernels added in the

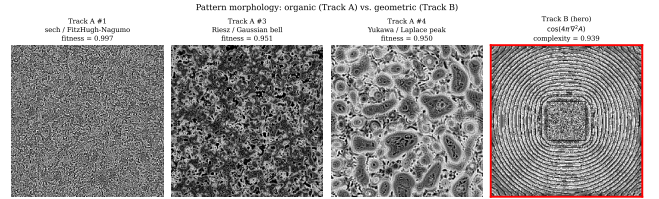


Figure 2: Morphological comparison between top Track A entries (organic) and the hero Track B expression (geometric).

final expansion to 28 account for 18%; crucially, 11 of these 13 new kernels produce archive entries. Top new kernels: Morse (7 entries), sech (7), Wendland C2 (6), Riesz (6). FitzHugh-Nagumo leads growth functions with 14 entries — notable as a neuroscience-derived excitable medium model. The four geometries contribute approximately equally (21–29%), suggesting self-organization is robust across geometric substrates. Per-family counts are in the released repository.

Localization and Locomotion Discovery

The parametric $G(K * A)$ form creates implicit smoothness constraints: 99.3% of Track A entries survive long evaluation, while Track B expressions frequently produce degenerate states. The main archive contains only extended textures; targeted searches with localization-rewarding fitness reveal

a progression toward localized structures.

Stationary solitons. With isotropic kernels, the `power_law/gaussian_bell` combination produces compact structures (~ 30 pixels, concentration 0.91) persisting 10,000+ steps without motion. This kernel is not in the original Lenia space.

Moving solitons. Breaking spatial symmetry via anisotropic kernel perturbation enables locomotion. A search over 14,000 genomes discovers **50 moving solitons** (net displacement > 20 pixels, concentration > 0.7 , alive fraction $< 15\%$) from **17 kernel/growth combinations** (Figure 4 shows six representatives). Of these, 49 come from expanded function families; one (`polynomial/sigmoid_pair`) uses a kernel from the original Lenia space paired with a new growth function, suggesting that novel kernel-growth *combinations* — not only novel kernels — can unlock locomotion. This does not imply original kernels cannot support locomotion — Chan (2019) demonstrated Orbium with Gaussian kernels through fine-tuned parameters. Rather, expanded function families appear to have **wider locomotion basins** accessible to coarse-grained evolutionary search (mutation of 10–30% of parameter range).

The most robust creature is a **bessel/gaussian_bell** soliton (Figure 3, top): a ~ 10 -pixel ring surviving all 9 trials across 3 initial positions $\times$ 3 random seeds (heading measured as the angle of net centroid displacement; $\text{std} = 2.1^\circ$ over the 9 trials), invariant to initial conditions and determined by the kernel’s anisotropy angle. The morphologically richest is a **morse/relu_like** creature (Figure 3, bottom) with tadpole-like asymmetric form and continuous shape deformation (6/9 survival, position-dependent direction).

Kernel family	Origin	Cands.	Best disp.
yukawa	Nuclear physics	14	56 px
sinc	Signal processing	6	173 px
riesz	Fractional calculus	3	119 px
elliptical	Geometry	3	93 px
morse	Molecular physics	1	75 px
lennard_jones	Molecular physics	4	55 px
polynomial*	Original Lenia	1	38 px

Table 4: Locomotion candidates by kernel family (selected). *Original Lenia kernel paired with new growth function (`sigmoid_pair`).

Within our coarse-grained search, anisotropic kernel perturbation reliably opened directed-motion basins, while the isotropic runs yielded only stationary solitons. We do *not* claim isotropy precludes motion: Chan’s isotropic-kernel Orbium glides through an *emergent* asymmetric pattern. Injecting anisotropy into the kernel instead hand-builds the symmetry breaking that standard Lenia discovers emergently — a less emergent but far more search-accessible route, giving expanded kernels **wider locomotion basins** reachable by coarse mutation. Locomotion concentrates in

oscillatory kernels (sinc, *bessel*) and long-range attractive-repulsive potentials (*yukawa*, *morse*).

Ablation: Random Expression Baseline (Track C)

Track C generates 1,000 expressions from a fixed context-free grammar (terminals `conv(A)`, `laplacian(A)`, `A`, and numeric constants; unary `sin / cos / tanh / exp / sigmoid`; binary `+, −, ×`), with *no* mutation or parent selection. Each passes the same compile-and-run validity gate as Track B and is scored into the same archive. It isolates what *un-guided* sampling achieves, separating the function space’s value from that of LLM guidance.

Metric	Track A	Track B	Track C
Evaluations	$\sim 125\text{K}$	$\sim 5\text{K}$	1,000 (809)
Archive bins	291	6	136
Best fitness	0.997	0.939*	0.967
Entries > 0.9	12	1	3

Table 5: Three-track comparison. *Controlled re-evaluation at 128^2 , 8,000 steps.

Three findings: (1) **Function space is inherently diverse** — 1,000 random expressions fill 136 bins, demonstrating behavioral diversity inaccessible to parametric search. (2) **Parametric bias contributes quality** — Track A wins 132/136 head-to-head fitness comparisons against Track C. (3) **LLM provides precision tuning** — Track C independently discovers $\cos(\nabla^2 A)$ in 6/136 entries but without the 4π frequency that creates spatial resonance with the kernel radius; the LLM’s mathematical knowledge enables precise tuning producing qualitatively novel crystallographic patterns.

Discussion

Structural Expressiveness and Decomposition

The discovery of $\cos(4\pi \cdot \nabla^2 A)$ reveals that the space of self-organizing mathematics is *structurally stratified*: the standard $G(K * A)$ separation between spatial interaction and local response precludes spatially-aware growth, and some mathematical regions are accessible only by expanding the framework’s expressiveness. The three-track ablation cleanly decomposes contributions: the function space provides diversity (136 bins from 1,000 random samples), the parametric template provides quality (winning 97% of head-to-head comparisons), and the LLM provides precision tuning of promising structural motifs. We therefore read the central contribution as the *framework expansion* — admitting differential operators into growth — more than the single LLM-found rule: Track C independently reaches $\cos(\nabla^2 A)$, so the expanded space makes the curvature-coupled *regime* reachable, with the LLM tuning within it. The practical insight for LLM-guided discovery

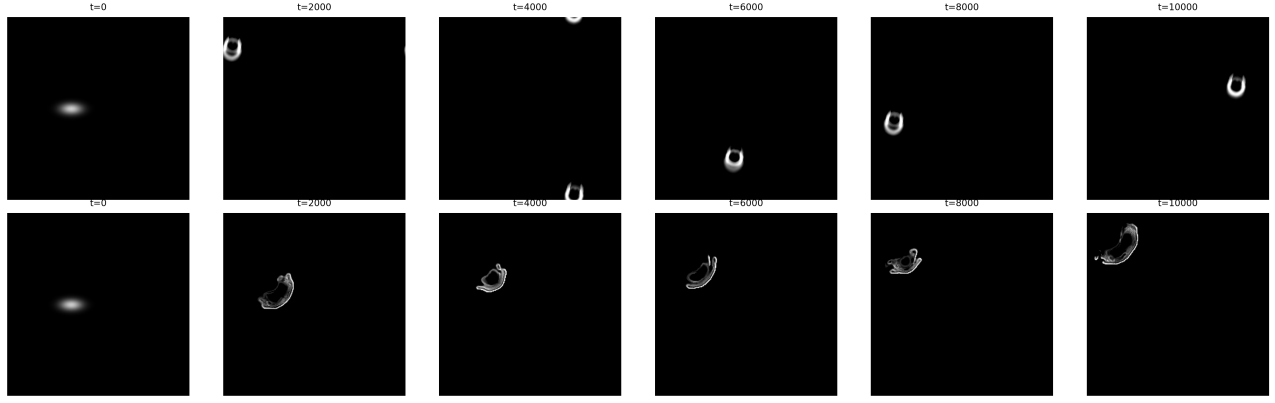


Figure 3: Moving solitons from expanded kernel families at 256×256 . Top: `bessel/gaussian_bell` — a ~ 10 -pixel ring with kernel-determined autonomous motion (9/9 robustness, direction std = 2.1°). Bottom: `morse/relu_like` — a tadpole-like creature with asymmetric form and continuous shape deformation (6/9 survival).

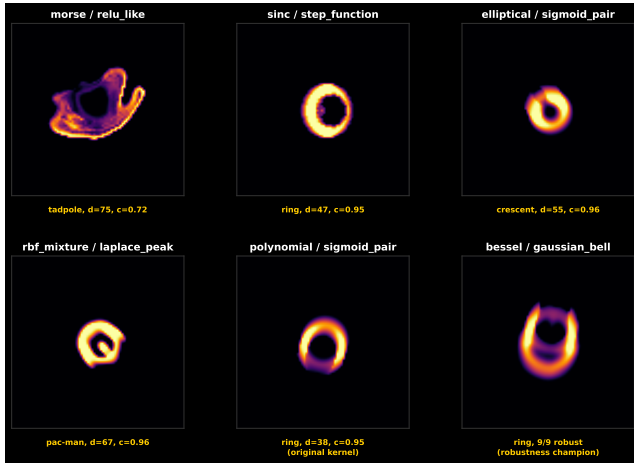


Figure 4: Gallery of 6 moving solitons from different kernel families. Top: `morse/relu_like` (tadpole, $d=75$), `sinc/step_function` (ring, $d=47$), `elliptical/sigmoid_pair` (crescent, $d=55$). Bottom: `rbf_mixture/laplace_peak` (pac-man, $d=67$), `polynomial/sigmoid_pair` (ring, $d=38$, *original kernel*), `bessel/gaussian_bell` (ring, 9/9 robust). Five use expanded kernels; one (polynomial) uses an original Lenia kernel paired with a new growth function.

is that domain-specific high-level operations yield substantially higher hit rates than abstract conceptual prompts.

From Textures to Locomotion

The progression texture $\rightarrow$ stationary soliton $\rightarrow$ moving soliton reveals a hierarchy: extended patterns require only a suitable kernel-growth combination; localization additionally requires appropriate spatial decay; and, under coarse search, locomotion was reached most readily by breaking symmetry in the kernel rather than waiting for it to emerge.

The moving solitons are small (~ 10 – 30 pixels) and simpler than Chan’s (2019) Orbium, arising from a different paradigm — coarse-grained search over expanded functions rather than fine-tuned parameters. Whether these represent fundamentally different locomotion mechanisms remains open.

Limitations

Our work has five main limitations: (1) only Lenia substrate — generalization to Gray-Scott, Particle Life remains untested; (2) unequal search budgets across tracks ($\sim 125K$ vs $\sim 5K$ vs $1K$ evaluations), only partially addressed by controlled re-evaluation; (3) Track B warm-start bias from Track A’s existing 287-entry archive; (4) high failure rate of LLM-generated expressions producing degenerate states; and (5) the per-cell fitness coincides with one behavior descriptor (complexity), so Track A’s archive reflects diversity illumination more than independent quality optimization, and a like-for-like ablation isolating MAP-Elites from random archive sampling remains future work.

Future Work

- **Cross-substrate transfer:** Test whether discovered expressions produce novel behaviors in other dynamical systems, demonstrating *universal self-organizing operations*.
- **Richer creatures:** Scale to Orbium-like complexity via higher resolution (512^2+), larger kernels, and multi-channel fields.
- **Equal-budget comparison:** Extend with matched compute across tracks at higher resolution.

Conclusion

We systematically expanded Lenia’s search space from 4 to 28 kernels, 4 to 16 growth functions, 4 geometric sub-

strates, and open LLM-generated expressions. MAP-Elites search across 125,000+ evaluations discovered 291 behavioral niches. LLM-guided search discovers $\cos(4\pi \cdot \nabla^2 A)$ — outside both single-kernel and additive multi-kernel Lenia — producing geometric patterns no standard parametric search reaches. A three-track ablation decomposes contributions from search space, inductive bias, and LLM guidance. Targeted localization search discovers 50 moving solitons from 17 kernel/growth combinations — 49 from expanded families (bessel, morse, yukawa, sinc), one from an original kernel with a new growth function — with anisotropic kernels widening the locomotion basins reachable by coarse search. The mathematical foundations of artificial life are broader than previously understood.

Code, data, archive files, and soliton animations are available at <https://github.com/memess2001/lenia-search>.

Appendix

A. Complete Kernel Function Definitions

Category	Functions	Mathematical origin
Classic (4)	Gaussian, polynomial, Mexican hat, bump	Original Lenia
Oscillatory (6)	Bessel, Gabor, sinc, cosine, chirp, Fourier series	Signal processing
Physics-inspired (6)	Lennard-Jones, Morse, Riesz, Yukawa, LoG, DoG	Molecular physics
Windowed (5)	Hann ring, Blackman-Harris, Wendland C2, plateau, sech	Approximation theory
Geometric (7)	Multi-ring, power-law, step ring, elliptical, spiral, Matérn, log-Gaussian	Spatial statistics

Table 6: Summary of 28 kernel function families.

B. Complete Growth Function Definitions

Category	Functions	Origin
Classic (4)	Gaussian bell, step function, asymmetric bell, bistable	Original Lenia
Heavy-tailed (3)	Laplace peak, Cauchy peak, ReLU-like	Robust statistics
Ecological (4)	Ricker, Allee effect, Gompertz, Brusselator	Population dynamics
Dynamical systems (3)	FitzHugh-Nagumo, Swift-Hohenberg, witch	Reaction-diffusion
Wavelet (2)	Sinc growth, sigmoid pair	Signal processing

Table 7: Summary of 16 growth function families.

C. Hero Expression and Component Analysis

The full LLM-generated expression (bin [19,19], complexity 0.939 at 128^2):

$$A_{t+1} = \text{clip}\left(A + 0.31 \cdot \left[\max\left(e^{-\frac{(\text{conv}(A) - (\mu_0 + 0.03 \cos(4\pi \nabla^2 A)))^2}{2 \cdot 0.055^2}}, \tanh(\dots)\right) \cdot W + G_2\right], 0, 1\right) \quad (2)$$

where $u = \text{conv}(A)$, $\mu_0 = 0.12$, the band-pass window $W = \text{threshold}(u + \nabla^2 A, 0.08) (1 - \text{threshold}(u + \nabla^2 A, 0.32))$, and the harmonic term $G_2 = 0.12 \sin(7\pi(u - 0.5\nabla^2 A)) e^{-((u - 0.16)/0.065)^2/2}$. (The exact machine-readable form is in the released repository.) The three components interact as follows. The *Laplacian-modulated growth* G_1 is a Gaussian bell whose center shifts with $\cos(4\pi \nabla^2 A)$; the cosine’s zero crossings discretize the Laplacian field into periodic stable bands, producing the concentric rings — the structural innovation, since standard Lenia’s $G(\mu, \sigma)$ has a fixed center μ . The *bandpass window* W confines growth to intermediate densities $0.08 \leq u \leq 0.32$ (preventing both extinction and saturation), and the *harmonic term* G_2 adds a secondary oscillation (~ 3.5 cycles) that creates sub-structure within the rings.

Expressibility, stated precisely: Single-kernel Lenia applies a single-argument growth G to one convolution $u = K * A$. (i) $\nabla^2 A$ is a distinct second-order linear operator not obtainable as $K * A$ for any *admissible* kernel: the discrete Laplacian is zero-mean ($\int K = 0$) and sign-indefinite, while Lenia kernels are non-negative and normalized to $\int K = 1$. (ii) Additive multi-kernel Lenia $\sum_i G_i(K_i * A)$ aggregates one-argument growths linearly and cannot reproduce the hero’s nonlinear coupling, in which $\nabla^2 A$ shifts the center

of a Gaussian bell evaluated at $\text{conv}(A)$. Hence the term lies outside the *single-kernel and additive multi-kernel* parametric families. This is a statement about those families, *not* a universal non-representability claim: multi-channel Lenia and Neural Cellular Automata, which expose multiple fixed convolutions to a joint cross-channel map, can represent it.

Top Track A entries for comparison: sech/FitzHugh-Nagumo (fitness 0.997), Riesz/Gaussian bell (0.972), and Yukawa/Laplace peak (0.950), all on flat plane geometry. FitzHugh-Nagumo — a neuroscience-derived excitable medium model — contributes 14 archive entries including 2 of the top 3.

D. Released Materials

The released repository (<https://github.com/memess2001/lenia-search>) contains the full LLM system, generation, and mutation prompt templates; per-component computational details (hardware, run-times, and costs); the prompt-strategy statistics (88% of Track B entries use high-level operations such as `curl/compete/conserv`, with hit rates rising from 0.4% to 13.7% under relaxed dual-field prompting); all archive files; and rendered animations of the discovered solitons, including the $\cos(4\pi \nabla^2 A)$ pattern and the `bessel/gaussian_bell` and `morse/relu_like` creatures.

Acknowledgments

We disclose the use of large language models (Claude, Anthropic) in this work. As a *method component*, Claude (Claude Sonnet 4, model `claude-sonnet-4-20250514`) generated the candidate update-rule expressions evaluated in Track B (Section 3.5; full prompts in the released repository); these generations are an object of study, not a writing aid. As a *writing and tooling aid*, the authors additionally used Claude to help draft and revise the manuscript prose and to assist in producing figure-rendering and analysis code. All experimental designs, results, mathematical claims, and the final text were verified by the authors, who take full responsibility for the content.

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